Stratigraphy and paleomagnetism of Brunhes and Matuyama (>790 ka) Quaternary deposits at Merritt, British Columbia[1]

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A succession of Quaternary deposits in the Merritt basin of south-central British Columbia contains evidence for four glaciations and two interglaciations. Paleomagnetic signatures in these sediments are of three types: normal polarity, proposed to have been acquired during the Brunhes Normal Polarity Chron; reversed polarity, proposed to have been acquired during the Matuyama Reversed Polarity Chron; and reversed polarity (Matuyama age) all but obscured by a normally magnetized overprint (Brunhes age).

Reversely magnetized deposits at the base of the succession include glacial lacustrine deposits, interpreted as representing two different glaciations, and a paleosol and a succession of nonglacial sediments, which are evidence of two interglaciations. As the reversed polarity of these deposits is proposed to have been acquired during the Matuyama Reversed Polarity Chron, they are older than 790 ka.

Normally magnetized deposits, which make up the rest of the succession, contain evidence for only two glaciations, but traces of other glaciations may have been removed during the erosion interval encompassed by a major unconformity that underlies deposits of the last glaciation. All are referred to the Brunhes Normal Polarity Zone. In addition to these glacial and interglacial deposits, a series of normally magnetized Quaternary basalt flows forms a bench 90 m above the floor of the basin. These basalts were extruded after 790 ka but before the penultimate glaciation.

L'étude d'une succession de dépôts quaternaires, dans le bassin de Merritt du sud-central de la Colombie-Britannique, a permis de reconnaître quatre événements de glaciation et deux événements d'interglaciation. Les signatures paléomagnétiques qui marquent ces sédiments sont de trois types : polarité normale supposément acquise durant la chronozone polaire normale de Brunhes, polarité inverse acquise présumément durant la chronozone polaire inverse de Matuyama et polarité inverse (âge de Matuyama) presque obscurcie par la surimpression d'une aimantation normale (âge de Brunhes).

Les dépôts caractérisés par une aimantation inverse à la base de la succession incluent des dépôts lacustres glaciaires, qui selon notre interprétation représentent deux glaciations différentes, et un paléosol et une succession de sédiments non-glaciaires qui indiquent deux interglaciations. Vu que les résultats de l'étude suggèrent que la polarité inverse de ces dépôts fut acquise durant la chronozone polaire inverse de Matuyama, ils sont plus vieux que 790 ka.

Dans les dépôts caractérisés par une aimantation normale, soit le reste de la succession, on reconnaît la preuve de seulement deux glaciations, mais les traces de d'autres glaciations ont pu être effacées durant l'intervalle d'érosion, révélé dans la région par une discordance majeure sous-jacente aux dépôts de la dernière glaciation. Tous sont rattachés à la magnétozone de polarité normale de Brunhes. En plus de ces dépôts glaciaires et interglaciaires, un série de coulées de basalte quaternaire, d'aimantation normale, forme un banc de 90 m au-dessus du fond du bassin. Le période d'extrusion de ces basaltes est postérieure à 790 ka mais antérieure à l'avant-dernière glaciation.

[Traduit par la rédaction]

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Introduction

This paper describes the stratigraphy and paleomagnetism of Quaternary deposits in the vicinity of Merritt, British Columbia, and outlines the succession of recorded Quaternary events. The Merritt basin, lying about 250 km east of Vancouver in the southern part of the Interior Plateau of British Columbia (Fig. 1), contains a succession of glacial, nonglacial, and volcanic Quaternary deposits that were formed during several glacial and nonglacial periods. These deposits are exposed in separated stratigraphic sections and generally contain neither organic nor other materials that could be used to determine their age. Within each section the order of inferred geological events and magnetic polarities is defined by superposition. By keeping the number of events to a minimum and by arranging the magnetic polarities in the simplest possible order (that is, an older reversed sequence followed by a younger normal sequence) the sections can be stacked in stratigraphic order. On this basis it has been possible to obtain a record of at least four glacial events that date back to before the Brunhes–Matuyama reversal. Following the interpretation of Richmond and Fullerton (1986), this report uses an age of 790 ka for the Brunhes–Matuyama reversal (see also Johnson 1982, Shackleton *et al.* 1990).

Many of the geological observations on which this paper is based were made by R. J. Fulton during mapping of the Quaternary geology of the area from 1960 to 1962 (Fulton 1975). A number of subsequent visits were made and important additional information gathered during the period 1986–1989. Samples for paleomagnetic analyses were collected by R. J.

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Fulton during 1986–1988 and by R. J. Fulton and E. Irving in 1989. Paleomagnetic analyses of samples were conducted at the Pacific Geoscience Centre by P. M. Wheadon and E. Irving. R. J. Fulton is responsible for the geological observations and interpretations and E. Irving and P. M. Wheadon for the analysis and interpretation of paleomagnetic results.

Setting

The Merritt basin is part of a broad valley located near the southern end of the Interior Plateau of British Columbia (Bostock 1948). Its floor lies at an elevation of about 600 m, and the adjacent rolling uplands reach elevations of 1000–1400 m (Fig. 2). The area is part of the dry belt of southern interior British Columbia, with open grasslands occupying valleys and lower uplands (Weir 1955) and forests covering the higher areas. The Nicola River, which rises to the east within the Interior Plateau, drains the Merritt basin westward to the Fraser River. The Coldwater River, which has its headwaters in the Cascade Mountains to the south, joins the Nicola River near Merritt (Fig. 1).

The Merritt basin is in the Intermontane Belt of Tipper *et al.* (1981) (see also Monger *et al.* 1982). Bedrock immediately adjacent to the basin consists of the Nicola Group, a series of middle Mesozoic marine volcanic rocks with minor associated sedimentary units (Cockfield 1948; Preto 1979). The Merritt basin itself is underlain by the Coldwater beds, a series of sandstones, shales, and conglomerate with seams of coal (Cockfield 1948). These have been correlated with the Eocene Tranquille Formation of the Kamloops Group (Rouse and Mathews 1961; Ewing 1981). To the south of Merritt the sequence contains hornblende dacite dated by the K–Ar whole-rock method at 49.2 Ma (Thorkelson 1989).

Sequences studies

The Quaternary deposits discussed in this report occur at four main localities in the vicinity of Merritt. The prime locality (Lily Lake Road locality, A of Fig. 1) consists of 1 km of road cuts on the Lily Lake Road about 3 km south of the bridge over the Coldwater River. The Coldwater locality (B of Fig. 1) consists of a low escarpment and a series of gullies 100 m west of the Coldwater River and about 4.5 km south of its confluence with the Nicola River. The Merritt locality (C of Fig. 1) is a cutbank of the Coldwater River directly downstream from the bridge leading to the Lily Lake Road. The fourth locality (D of Fig. 1) is west of an abandoned quarry immediately south of Chutter Ranch, 7.5 km east of Merritt. Paleomagnetic data from a locality in Quilchena Creek valley 19 km east of Merritt (E of Fig. 1) and from several other localities in the region are also briefly discussed.

Lily Lake Road locality

Sections on and adjacent to the Lily Lake Road expose about 60 m of stratigraphic succession (Fig. 3). The lowest unit in this succession is a glacial lacustrine deposit here referred to as the sub-Coutlee silts (unit 2 of Table 1). The beds are tilted 20–25°, are about 10 m thick, consist of grey silt and clay, contain no organic remains, and have the same general appearance as glacial lacustrine sediments formed during the last glaciation. Couplets (varves?) are poorly developed, and several beds consist of redeposited silt and clay rip-up clasts. These observations, combined with the fact that there is no consistent trend in rhythmite thickness through the section (a characteristic shared with many of the glacial lacustrine deposits in the area that directly underlie the surface till but with only a few of those that overlie the surface till), suggest that the unit was deposited near an ice front during a glacial advance. The exposure of sub-Coutlee silts is at the bottom of a landslide scar, so it is likely that the 20–25° tilt of the beds is due to recent slumping.

The sub-Coutlee silts are overlain unconformably by up to 40 m of red-brown sand, silt, and gravel referred to as the Coutlee sediments (Fig. 3; unit 3 of Table 1). The lowest part of the Coutlee sediments is exposed only in a slump block, and its contact with the sub-Coutlee silts is poorly exposed. As noted in Table 1, this unit contains fossil mollusc shells and a tephra (the Coutlee tephra of Westgate and Fulton 1975; Fig. 4*a*). A thermoluminescence age of >670 ka has been obtained for the tephra by Berger (1985). The regularly bedded fluvial sediments (Fig. 4*b*) contain lenses of colluvium (Fig. 4*c*). Gravel occurring in the lower third contains mainly rounded clasts commonly of nonlocal (granitic and plutonic) lithologies. Gravel in the remainder of the succession contains mainly angular clasts consisting of locally derived pink rhyolite. Although there is considerable variation in texture throughout the succession, the sediment is generally coarser and better sorted in the lower part of the section than nearer the top, where it consists largely of silt and sandy silt. Finely disseminated iron oxide occurs throughout the Coutlee sediments. The paleosol part of this succession (Fig. 3) is a complex of soil horizons developed in loess, alluvial–lacustrine silt and sand, and colluvium. Features identifying this as a soil are downward gradation from partially leached A horizons to B horizons containing clay skins indicating the presence of eluvial clay; the presence of ped structures with black iron–manganese stains on vertical fractures; occurrence of C horizons highlighted by small lenses, vertical veinlets, and nodules of secondary calcium carbonate; and marginally brighter colours in the layers identified as B horizons. The generally poor development of individual soil horizons and the complex stacking of horizons developed in a variety of parent materials indicate periodic sediment deposition during soil formation. The brown colours of B horizons and the moderate degree of translocation of clays suggest that if the paleosol was on the present surface, it would be classified as a brunisol (Canada Soil Survey Committee 1978). The combination of these features and the characteristics of the Coutlee sediments indicate that the Coutlee sediments were deposited during a long nonglacial period, which was probably an interglaciation.

The beds above the Coutlee sediments consist of proglacial, interstratified, grey-coloured sand, gravel, and silt overlain by diamicton. These are referred to as the Kamloops Lake Drift (Fulton and Smith 1978; unit 6 of Table 1). The bottom of the unit sits on a prominent, angular unconformity marked by a discontinuous lag of boulders (<2 m in diameter) and coarse gravel (Fig. 4*d*). The boulders are of a variety of compositions; granitic lithologies are dominant, but argillite, volcanic agglomerate, trachyte, and fresh-appearing, vesicular, diktytaxitic olivine basalt (similar to valley basalt as described below) are also present. The interstratification of lithologies within this unit of the Kamloops Lake Drift signifies abrupt changes in the sedimentation during deposition. Imbrication of pebbles in the gravel suggests north to south transport, opposite to the flow direction of the present Coldwater River. In

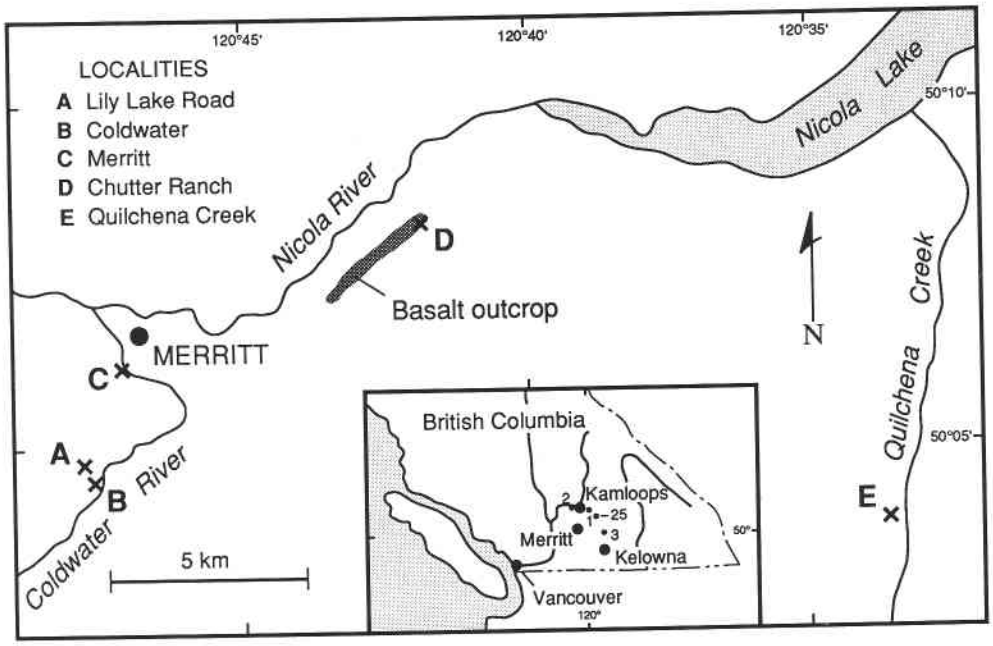


FIG. 1. Location map.

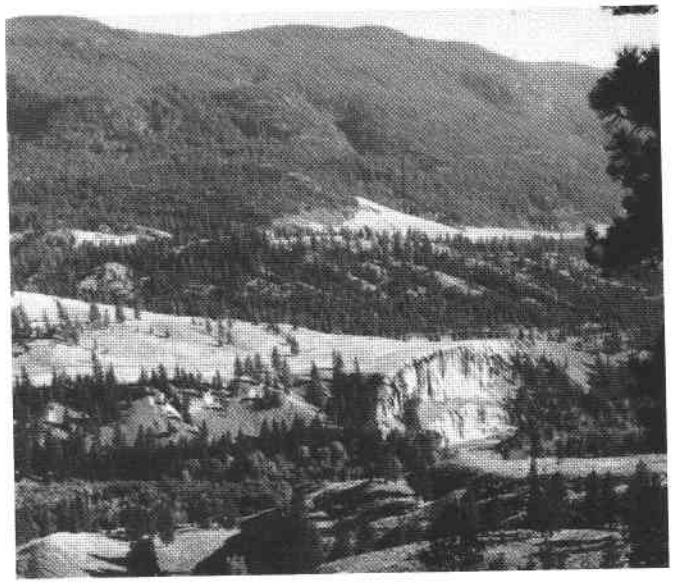


FIG. 2. General view across the Coldwater River valley from the Lily Lake Road locality. Note the 30 m exposure of till in the lower right corner of the figure (GSC no. 205436B).

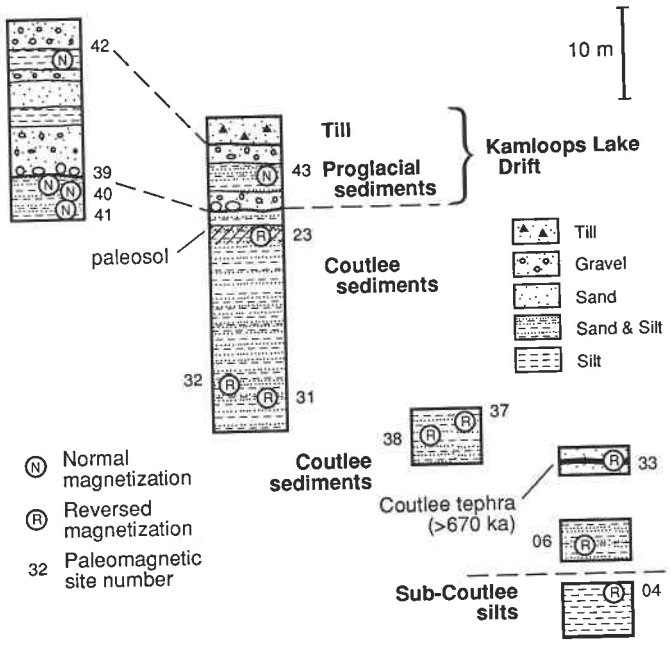


FIG. 3. Stratigraphic diagram, Lily Lake Road locality (A of Fig. 1). The sites with reversed polarity include those with both R and R_s magnetization.

most places these proglacial sediments are capped by till, and are folded and faulted with slices incorporated into the overlying diamicton.

Diamicton of the Kamloops Lake Drift in the Lily Lake Road succession is 2–6 m thick (unit 7 of Table 1), contains some striated clasts, includes sheared upward slices of the underlying sediment, and is interpreted as a till. Most pebbles in the till are rounded (derived from underlying fluvial gravels), and include fresh-appearing, vesicular, diktytaxitic olivine basalt (similar to valley basalt). In most places the till is massive or partly stratified. On the opposite side of the Coldwater River, 750 m to the east, the till is moderately well stratified and 30 m thick (Fig. 2).

At the top of the Lily Lake Road succession (100 m south of the successions illustrated in Fig. 3) the till is overlain by 1–2 m of laminated grey silt and clay that are part of the Upper Stratified Kamloops Lake Drift (Fulton and Smith 1978), and in this report are referred to as Merritt silts (unit 8 of Table 1). Individual rhythmites (varves?) are from 0.01 to 0.15 m thick and grade thinner upwards. The elevation at which this sediment occurs suggests that they were deposited in glacial Lake Hamilton (Mathews 1944; Armstrong and Fulton 1965).

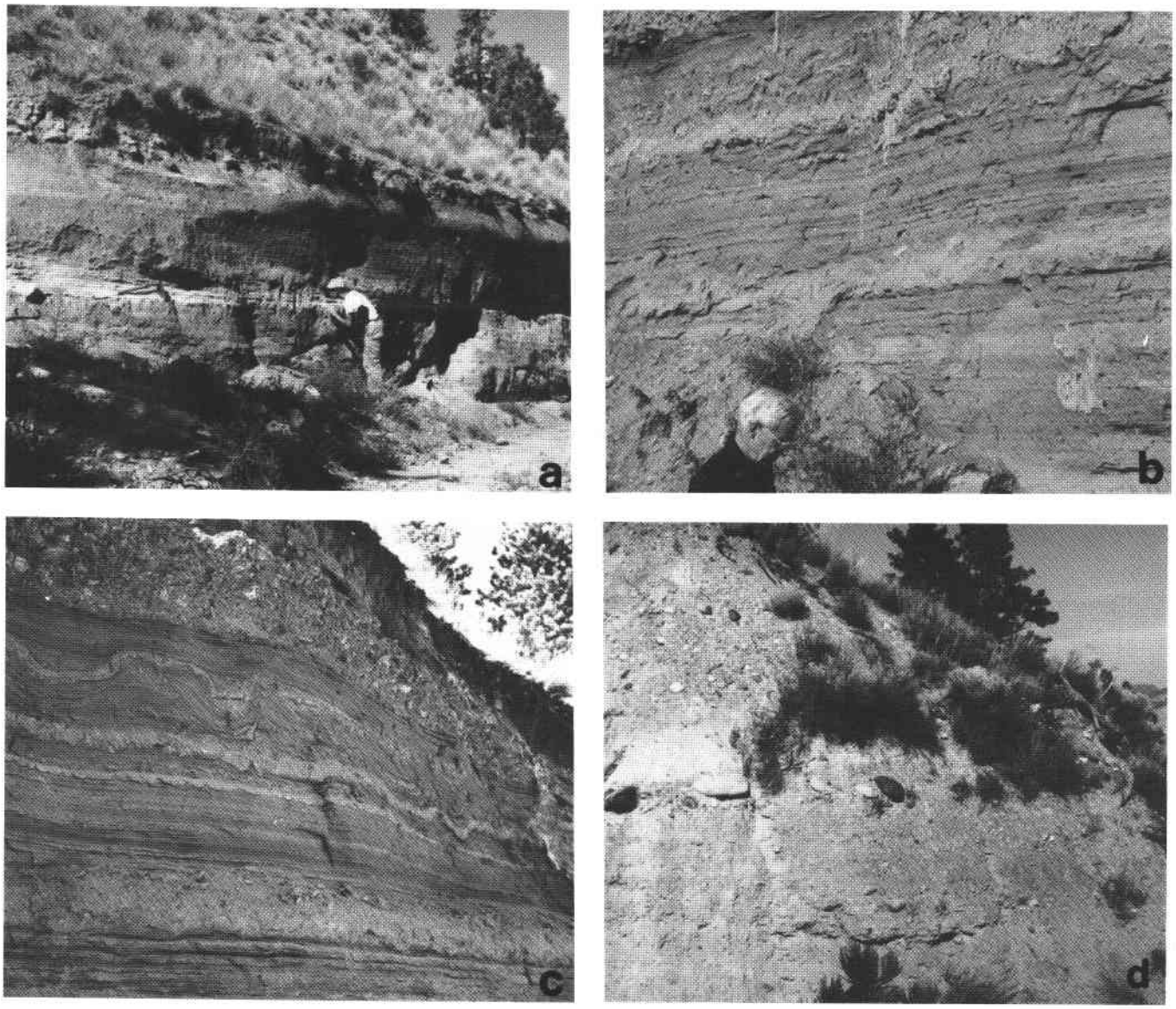


FIG. 4. Coutlee sediments. (*a*) Tephra bed (GSC no. 116588). (*b*) Regularly bedded sand and silty sand (GSC no. 205434). (*c*) Penecontemporaneous deformation features in sand and silty sand—note talus-like gravel near the top of the exposure (GSC no. 116586). (*d*) Boulder lag at the top of the Coutlee sediments (GSC no. 205436A).

Coldwater locality

The Coldwater locality is on the west side of the Coldwater River about 400 m east of the Lily Lake Road (B of Fig. 1). Except for the Merritt silts and till of the Kamloops Lake Drift at the top, the units are different from those at the Lily Lake Road locality (Fig. 5).

The lowest unit at this locality, termed the Coldwater silts, is probably the oldest Quaternary deposit described in this paper (unit 1 of Table 1). The Coldwater silts are interpreted as glacial lake sediments, and are similar in stratification and texture to sediments deposited in this region at the end of the last glaciation (Fulton 1965). They differ from the sub-Coutlee silts (unit 2), in being better stratified, being coarser grained, and consisting of more sharply defined couplets that grade finer and thinner upwards. At least 30 couplets (varves?) were counted, with many thin couplets that were too disrupted for counting in the upper 2.5 m of the unit. This disrupted zone consists of fractured and weathered silts and is interpreted as representing a soil. The yellow colour of this paleosol is stronger than that developed in similar materials during Holocene time, so the paleosol is interpreted as having developed during an interglaciation that was as warm as, and probably longer than, the present one. The Coldwater silts are tilted 20° towards the west. This might indicate that they are part of a block that slumped down from the vicinity of the Lily Lake Road locality. This seems unlikely however, because similar sediments are not exposed at the Lily Lake Road locality. The unit overlying the Coldwater silts is interpreted as a subglacial deposit, so the tilting of the silts might be due to glacial tectonism. Rotation of the magnetic declinations in the Coldwater silts (see below) is consistent with this hypothesis.

The complex of brownish-coloured contorted beds and lenses of gravel, diamicton, sand, and silt that overlies the Coldwater silts is referred to as brown drift (Fig. 5; unit 5 of Table 1). The diamictons in this unit are compact and clay rich, and are intimately interfolded with sands and gravels. In this discus-

TABLE 1. Quaternary stratigraphic units exposed in the vicinity of Merritt, British Columbia

Unit no.	Unit name	Locality	Thickness (m)	Lithology	Interpretation	Polarity	Age
8	Merritt silts (Kamloops Lake Drift)	Merritt Lily Lake Road Coldwater	>10	Grey laminated silt with minor fine-grained sand and clay; base is not exposed and top is present surface; laminations in couplets of alternating thin dark grey (clay rich) and thicker light grey (coarser grained) beds; couplet thickness ranges from 2.5 m at base to 0.01–0.02 m at top but there is considerable thickness variation from couplet to couplet; about 150 couplets were counted but some at the top have been destroyed by soil-forming processes	Glacial lake sediments	Normal	±11 ka
7	Till (Kamloops Lake Drift)	Lily Lake Road Coldwater	2–6	Grey-brown, stony diamicton with a sandy to silty matrix; base unconformably overlies "Proglacial sediments" and top is either overlain by thin laminated silts equivalent to the Merritt silts or underlies the present surface; large clasts mainly rounded and some are the same lithology as valley basalts; contains scattered sand lenses; contorted lenses of underlying stratified sediment are present near base of unit	Glacial deposit	No data	11–20 ka
6	Proglacial sediments (Kamloops Lake Drift)	Lily Lake Road	<17	Sequence of grey, unoxidized sand, gravel, and silt units; gravel at base with boulders <2 m on erosional unconformity (some of these are same lithology as valley basalts); top folded and faulted at base of overlying till; succession of units at one place, from base to top: 6 m of coarse bouldery gravel with imbrication dipping north, 2 m of horizontally stratified sand and silt, 3 m of horizontally stratified and ripple-laminated sand, 1 m of pebbly gravel, 2.25 m of silt and sandy silt with indistinct horizontal bedding, 1 m of massive pebble gravel, 1.5 m of massive medium- to coarse-grained sand	Proglacial deposits	Normal	±20–25 ka
5	Brown drift	Coldwater River	<8.5	Series of brownish-coloured lenses and contorted beds of gravel, diamicton, sand, and silt; at base this unit blends into underlying weathered and disrupted laminated silts (Coldwater silts); at top beds are deformed so they parallel the undulating contact of the overlying grey-coloured glacial complex (Kamloops Lake Drift); succession at one place: 6 m of sand and gravel with silt at top, 0.5 m of diamicton, 1 m of gravel, 3 m of clayey silt diamicton, 1 m of gravel	Glacial deposits	Normal	>25 to <790 ka
4	Valley basalts	Chutter Ranch Quilchena Creek valley	>20	Series of irregular, interleaving flows of blocky, light grey, olivine basalt; up to five flows were seen in a single exposure; base is not exposed and top is the present surface (glacially striated); flows are <5 m thick, massive in centre, and vesicular, glassy, and ropy within 0.5 m of flow tops and bottoms; minor unoxidized silt coats ropy lava, but in general there is no evidence of interflow sediments or paleosols; according to Lambert (1963), the basalt consists on average of 45% feldspar (labradorite), 28% glass, 14% pyroxene (pigeonite and augite), 10% olivine, and minor magnetite, ilmenite, and hematite; texture is diktytaxitic	Volcanic eruption	Normal	0.6 ± 0.4 Ma (Church 1980) 100 to <790 ka

TABLE 1 (*concluded*)

Unit no.	Unit name	Locality	Thickness (m)	Lithology	Interpretation	Polarity	Age
3	Coutlee sediments	Lily Lake Road	<40	Interstratified, red-brown sand, gravel, silt, and rubble; base rests unconformably on grey silt; top underlies an unconformity that is overlain by a bouldery gravel; brownish rusty colour due to finely disseminated iron oxide, which coats many of the clasts; well-sorted sand beds locally cemented by calcium carbonate; basal 16 m consists of sand and gravel with minor silt and includes a 0.15 m tephra bed with a thermoluminescence age of >670 ka (Berger 1985); the gravels contain mainly rounded pebbles of distant provenance; fossils of *Anadonta* sp., *Valvata* sp., *Lymnaea* sp., and *Gyraulus* sp. occur adjacent to the tephra; the upper 24 m is dominantly horizontally stratified sand and silty sand with interbeds of rubble dominated by platy fragments of local pink rhyolite; massive to indistinctly stratified silty sand and sand containing thin lenses of rubble lie at the top; a 3 m thick complex containing several weakly developed paleosols occurs 1.75 m below the top at one site; paleosol horizons are dark brown and contain some translocated clay in upper part; near base of the dark-coloured horizons are thin calcium carbonate veins and lenses, and small nodules	Interglacial basin fill deposits	Normal at top but the rest reversed	~790 ka and >790 ka
2	Sub-Coutlee silts	Lily Lake Road	>10	Grey silt with minor fine-grained sand and clay; base not exposed; unconformably underlies Coutlee sediments; bedding dips 20–25° towards the adjacent valley; poorly developed couplets 0.1–0.25 m thick; at least 15 couplets were noted but exposure is very poor, contains calcium carbonate concretions and lenses	Glacial lake deposits	Reversed	>790 ka
1	Coldwater silts	Coldwater	>15	Grey laminated silt with minor fine-grained sand and clay; base not exposed; paleosol at the top unconformably underlies the "brown drift", bedding is tilted 20° away from the adjacent valley wall; couplets are well developed and consist of dark-coloured (clayey) units, generally <0.01 m thick, and light grey (silt and fine-grained sand) units that near base of the exposure are 0.2–0.5 m thick, and near the top are 0.05 m thick (several couplets in the middle are 1–2 m thick); a 2.5 m thick paleosol at the top of the unit consists of finely fractured, highly disrupted couplets of silt and clay modified to light tan and brownish yellow colours	Glacial lake deposits	Reversed	>790 ka

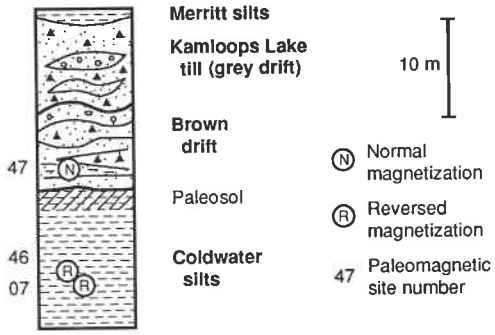


FIG. 5. Stratigraphic diagram, Coldwater River locality (B of Fig. 1). See legend in Fig. 3 for meaning of patterns.

sion the brown drift is interpreted as a separate glacial unit, but it equally well could be a complex of glacially tectonized sediments making up the basal part of the overlying Kamloops Lake Drift.

The brown drift is overlain by a grey sediment complex that consists of diamicton with a compact matrix and lenses and beds of sand and gravel. This is interpreted as a glacial succession and is correlated with the surface till found throughout this area (unit 7 of Table 1). A discontinuous veneer of stratified silt (correlated with unit 8 of Table 1) overlies the grey drift (Fig. 5).

Merritt locality

A bank that has been cut by the Coldwater River at the southern outskirts of Merritt provides an excellent exposure of the youngest unit discussed in this paper (C of Fig. 1; Fig. 6). This grey, rhythmically bedded silt, which constitutes the surface unit throughout much of the Merritt basin, is the uppermost unit of Kamloops Lake Drift (Fulton and Smith 1978). In this paper it is referred to as Merritt silts (unit 8 of Table 1). These deposits contain many features common to glacial lake deposits in this region: grain size fines upwards; the stratification consists of couplets made up of thick silt beds overlain by thin, more-clay-rich beds; couplets thin upwards; and thicker beds are made up of several graded units (Fulton 1965). The base of Merritt silts is not exposed at this locality, but in the sections described above they overlie the till of Kamloops Lake Drift.

Chutter Ranch locality

The Chutter Ranch locality lies at the top of the west-facing escarpment formed by the basalt flows that occupy the eastern part of the Merritt basin (D of Fig. 1). At least 15 m of basalt is present and the top of the escarpment is about 90 m above the present floor of the basin. These, and similar flows that occur near valley bottoms in this part of British Columbia, are referred to as valley basalts (Cockfield 1948; Lambert 1963; unit 4 of Table 1). The base of the basalts is not exposed at this locality, but farther to the south they overlie unconsolidated brown sand and gravel. Church (1980) reported a K–Ar age of 0.6 ± 0.4 Ma for this unit. Samples were also collected from basalt of similar appearance in Quilchena Creek valley to the east (locality E of Fig. 1).

Localities outside Merritt basin

Samples for paleomagnetic analysis were collected from various Quarternary units at several other localities in south-central British Columbia. Sites outside the Merritt basin are listed as a group in Table 2 and locations are shown on Fig. 1. These are included to provide additional information on the location of the paleomagnetic pole.

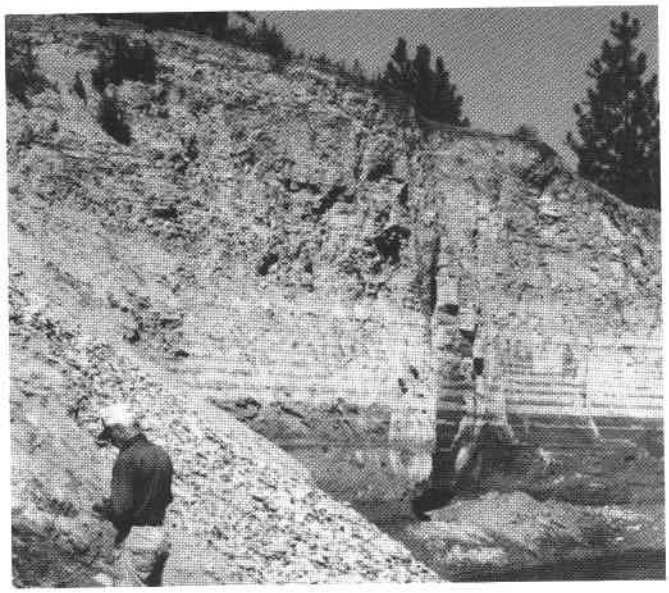


FIG. 6. Merritt silts exposed in a cutbank of the Coldwater River (GSC no. 205435).

Barnhart Vale

This locality (stop 6-1 of Fulton and Halstead 1972) is on the road to Barnhart Vale, 12 km east of Kamloops (1 on inset map, Fig. 1). Samples were collected from near the top of a cliff in the South Thompson silt (Fulton 1965). Like the Merritt silts, the South Thompson silt is a glacial lacustrine deposit laid down at the end of the last glaciation. Because the two localities are only about 70 km apart and deglaciation was relatively rapid in this area (Fulton 1969), the South Thompson silt is roughly contemporaneous with the Merritt silts.

Mission Flats

This locality is a sanitary landfill site, 6 km west of Kamloops (2 of inset map, Fig. 1). A site adjacent to that sampled is illustrated by Fulton and Halstead (1972, Fig. 8). Samples were taken from a grey clay (slough bed of Fulton and Halstead 1972) lying about 2 m below the Kamloops Lake tephra, which is slightly older than 34 ka (Westgate and Fulton 1975). This age, combined with the suggestion that the paleoclimate was cooler and more moist than present, led Fulton *et al.* (1987) to refer the sediments to Middle Wisconsinan.

Okanagan Centre

A gravel pit on the east side of Okanagan Lake about 15 km north of Kelowna (3 of inset map, Fig. 1) contains a succession of units that have been related to two glacial and two nonglacial periods (Fulton and Smith 1978). The samples analyzed were collected from rhythmically bedded (varved?), glacial lake silt and clay that make up the uppermost unit of deposits ascribed to the penultimate glaciation. These sediments are referred to the penultimate glaciation because they are overlain by eolian sand containing a paleosol and a tephra that are considered to be Middle Wisconsinan (Westgate and Fulton 1975), and that in turn are overlain by till and glaciofluvial sediments referred to the Fraser Glaciation, which is Late Wisconsinan (Fulton and Smith 1978).

TABLE 2. Paleomagnetic data

	Site	Lat. (N)	Long. (W)	UTM (E/N)	s	M (A/m)	D (°)	I (°)	k	α_{95} (°)
Kamloops Lake Drift										
45 *N*	Merrit silt (grey)	50°6.3′	120°47.8′	6579/55524	5	0.74	355	35	117	7
44 *N*	Merritt silt (grey)	50°6.3′	120°47.8′	6579/55524	8	0.87	348	35	112	5
42 *N*	Proglacial silt (grey)	50°4.6′	120°48.6′	6568/55492	7	0.43	341	75	8	23
43 *N*	Proglacial silt (grey)	50°4.8′	120°48.5′	6568/55495	6	0.26	341	48	16	17
Mean	4 sites						348	48	18	22
Valley basalts										
15 *N*	Quilchena Valley (highest)	50°4.2′	120°31.8′	6769/55491	5 (10)	1.90	002	68	3220	1
14 *N*	Quilchena Valley	50°4.2′	120°31.8′	6769/55491	4 (07)	1.24	014	67	758	2
13 *N*	Quilchena Valley	50°4.2′	120°31.8′	6769/55491	4 (08)	0.78	018	63	263	3
12 *N*	Quilchena Valley (lowest)	50°4.2′	120°31.8′	6769/55491	6 (12)	0.86	042	59	153	4
Mean	Quilchena Valley 4 sites						021	65	92	10
36 *N*	Chutter Ranch (highest)	50°8.2′	120°41.5′	6653/55562	7 (14)	2.42	024	79	1010	1
35 *N*	Chutter Ranch	50°8.2′	120°41.5′	6653/55562	7 (14)	4.12	041	76	179	3
34 *N*	Chutter Ranch (lowest)	50°8.2′	120°41.5′	6653/55562	7 (13)	2.00	041	72	222	2
Mean	Chutter Ranch 3 sites						037	76	378	6
Mean	Valley basalts 7 sites						026	70	80	7
Uppermost Coutlee sediments										
39 *N*	Silt	50°4.6′	120°48.6′	6568/55492	7	0.18	348	63	249	4
40 *N*	Silt	50°4.6′	120°48.6′	6568/55492	5	0.30	357	60	177	6
41 *N*	Silt	50°4.6′	120°48.6′	6568/55492	7	0.19	336	63	93	6
Mean	3 sites						347	62	239	8
Coutlee sediments										
23 *R*	Brown paleosol	50°4.8′	120°48.5′	6568/55495	15	0.03	176	−61	19	9
32 N_o	Reddish pebbly sand	50°4.8′	120°48.5′	6569/55496	4	0.07	315	72	50	13
32 R_s					4	Scattered reversed magnetizations				
31 N_o	Brown pebbly sand	50°4.8′	120°48.5′	6569/55496	6	0.05	003	68	97	7
31 R_s					3	Scattered reversed magnetizataions				
37 N_o	Brown silt	50°4.8′	120°48.5′	6570/55496	4	0.03	332	71	98	9
37 R_s	Brown silt				4	0.01	207	−56	73	11
38 N_o	Red-brown silt	50°4.8′	120°48.5′	6570/55496	5	0.10	325	75	96	4
38 R_s					6	Scattered reversed magnetizataions				
33 N_o	Brown sand	50°4.9′	120°48.5′	6570/55497	7	0.03	356	71	79	7
33 R_s					2	Scattered reversed magnetizations				
Sub-Coutlee silts										
06 N_o	Brown silt	50°4.9′	120°48.5′	6570/55497	4	0.05	347	69	26	18
06 R_s	Brown silt				10	Scattered reversed magnetizations				
04 *R*	Grey silt with concretions	50°4.8′	120°48.3′	6572/55497	27	0.13	203	−54*	16	4
Brown drift										
47 *N*	Brown sand	50°4.6′	120°48.4′	6573/55492	7	0.35	020	59	91	6
Coldwater silts										
07 *R*	Grey silt	50°4.6′	120°48.4′	6573/55492	16	0.22	250	−51*	36	6
46 *R*	Grey silt	50°4.6′	120°48.4′	6573/55492	6	0.14	238	−53*	44	5
Sites outside Merritt basin										
01 *N*	Barnhart Vale	50°39.7′	120°10.6′	6994/56162	8	0.36	358	59	202	4
02 *N*	Mission Flats	50°41.4′	120°26.8′	6803/56182	3	0.01	308	65	33	22
03 *N*	Okanagan Centre	50°02.4′	119°26.2′	3254/55457	8	0.03	333	59	6	25
25 *N*	Westwold	50°27.8′	119°46.9′	3025/55935	15	0.04	005	66	71	5
Mean	4 sites						341	64	46	14

NOTES: Sites are located by latitude and longitude and grid reference; s is the number of individually oriented specimens, except in the basalts, for which the number of oriented cores followed (in parentheses) by the number of specimens cut from them is given; M is the mean intensity of magnetization of the s specimens; D and I are declination and inclination of the mean direction relative to present horizontal where beds are flat lying (inclinations marked * are relative to bedding (site 04 dips 25° at 154°E of north; sites 07 and 46 dip 16° at 325°E of north)); k is Fisher's estimate of precision; and α_{95} the radius of the circle of confidence ($P = 0.05$). See Appendix for analytical details.

Westwold

This locality is a gravel pit located on the Douglas Lake Road just south of the Kamloops–Vernon highway about 45 km southeast of Kamloops (25 of inset map, Fig. 1). The unit sampled is an approximately 2 m thick marl that lies on top of a nonglacial gravel and underlies deposits of at least two glaciations. Fulton and Smith (1978) suggested that this unit was deposited at the end of the Sangamon Interglaciation, but because there are no absolute ages and the sequence above is poorly exposed, this unit could be older. As noted below, the samples are normally magnetized, so the deposit is assumed of Brunhes age and hence is <790 ka old.

Paleomagnetic data

Most examined outcrops consisted of uncemented sediments. Specimens were obtained by excavating vertical faces from which cubes were carved and then inserted into 2 cm cubical plastic holders. A few core specimens from concretions were obtained with gasoline-powered drills. Orientations were measured by magnetic and sun compasses. Several specimens, usually between four and eight, were collected at 30 selected levels or sites. At each site, specimens spanned a stratigraphic thickness of about 1 m. The stratigraphic positions of sampling sites are shown in Figs. 3 and 5.

Measurements were made on automated Schonstedt spinner magnetometers that allowed batches of 15 specimens to be processed unattended. Stepwise alternating-field (AF) demagnetization up to 100 mT was carried out using a Schonstedt GSD-5 demagnetizer. For fields up to 175 mT, a Stupavsky SI4 instrument was used. All demagnetizations were made sequentially along three axes.

Three different modes of behaviour were found. In the first (designated N) the principal magnetization is of normal polarity; we shall argue that this was acquired at or soon after deposition during the Brunhes Normal Polarity Chron. The second type (R) has a principal magnetization of reversed polarity; we shall argue that this was acquired in the Matuyama Reversed Polarity Chron. In the third type, the principal magnetization is a normally magnetized overprint (N_o), which all but obscures a subordinate magnetization of reversed polarity (R_s); we shall argue that N_o and R_s are of Brunhes and Matuyama age respectively.

The first type of magnetization (N) is illustrated in Fig. 7. The decay of intensity during demagnetization is smooth, and, not uncommonly, a substantial part remains after treatment at 175 mT. Intensities fall by an order of magnitude during demagnetization. Small components, probably viscous remanent magnetizations (VRM) imposed by the present field, are removed in low fields, but thereafter the orthogonal plots show linear decay to the origin, proof that a single normally polarized magnetization (N) dominates. There is no evidence that N is other than a record of the ambient field at the time of deposition.

Samples with N magnetizations were collected from Kamloops Lake Drift (four sites), the uppermost levels of the Coutlee sediments (three sites), brown drift (one site), valley basalts (seven lava flows at two localities), and four sites outside the Merritt basin, a total of 19 sites. Individual specimen directions in sediments with N magnetizations are plotted in Figs. 8 and 9. Individual specimen directions in valley basalts are plotted as dots in Fig. 10. Site mean directions are listed in Table 2 and plotted in Figs. 9–11.

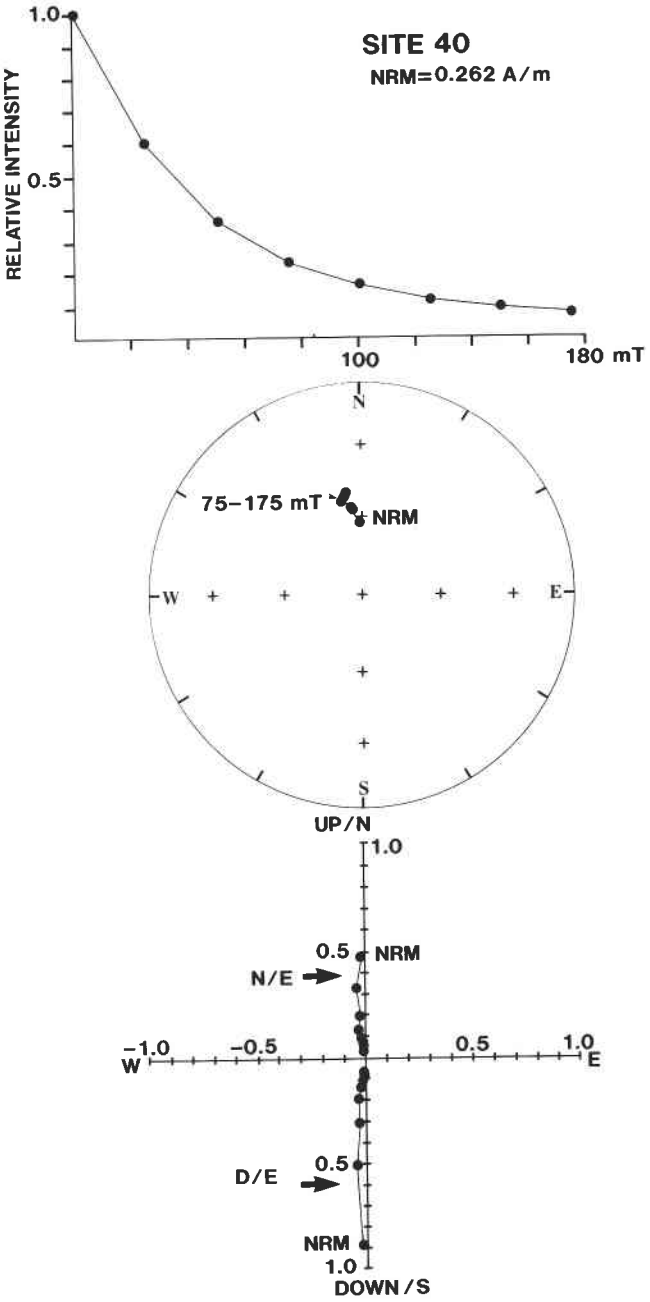


FIG. 7. Normal magnetization (N) in silt from uppermost Coutlee sediments, site 40. The beds are horizontal. Above, decay of intensity with increasing alternating magnetic field. Middle, changes in direction of magnetization during demagnetization. Below, two superimposed orthogonal plots in the down/east (D/E) and north/east (N/E) planes.

Examples of the second type of magnetization (R) are illustrated in Figs. 12 and 13. The first example is from the paleosol at the top of the Coutlee sediments (site 23). At first, the intensity increases, during demagnetization as a small present field VRM is removed. Thereafter, the magnetization decays smoothly, 40% remaining after treatment at 175 mT. The remaining, very stable principal magnetization is reversed in polarity. The orthogonal demagnetization plots show linear decay to the origin, indicating that only a single magnetization is present. A slightly different situation is illustrated by grey

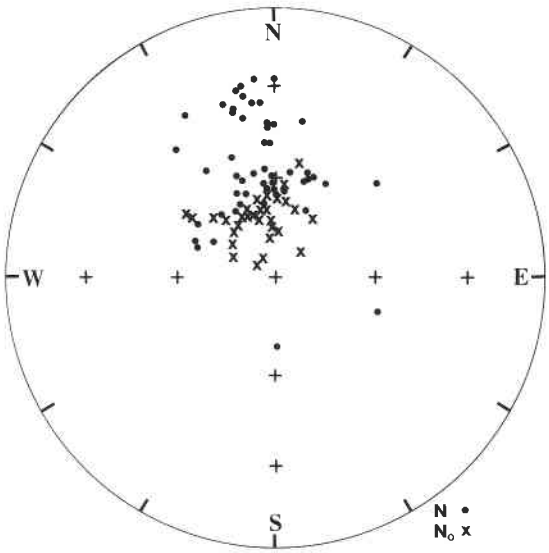


FIG. 8. Normal magnetization directions observed from sediments in the Merritt basin. N (52) are from specimens with Brunhes-age magnetizations. N_0 (30) are Brunhes-age overprints.

glacial lacustrine sediment at site 46 (Coldwater silts), which shows similar demagnetization behaviour, but the reversed magnetizations are directed towards the west-southwest (Fig. 13), the declinations being clockwise of those at site 23. Again, after removal of small present Earth's field VRM, the orthogonal plots decay linearly to the origin. These two examples demonstrate the existence of reversed magnetizations (R) in Coutlee sediments and Coldwater silts. There is no evidence that these R magnetizations are overprints, or that they are anything other than records of the geomagnetic field, acquired at the time the paleosol at site 23 formed (Fig. 12) and at the time of deposition of the lake sediment of site 46 (Fig. 13). Comparable R magnetizations occur also in the sub-Coutlee silts (site 04). Individual directions of specimens with R magnetization are plotted in the upper half of Fig. 14 and site mean directions are plotted in Fig. 11.

The third type of magnetization is shown in Fig. 15. The intensity decays rapidly during demagnetization, corresponding to a normally magnetized overprint (N_0) of low coercivity with a steeply inclined downward direction (labelled RM in Fig. 15). At high fields, a reversed magnetization (R_s) is revealed. R_s magnetizations occur at sites 06, 31, 32, 33, 37, and 38. Generally, the directions of R_s can be estimated only crudely, from ill-defined end points. They are plotted in the lower half of Fig. 14 and are widely dispersed. Site 37 is somewhat exceptional; although an N_0 magnetization of appreciable magnitude is present, R_s is well-defined (Table 2), and for this reason it has been included with R magnetizations in Figs. 11 and 14. All estimates of N_0 are plotted in Fig. 8. Site means are plotted as crosses in Fig. 11. The overall mean of N_0 magnetizations does not differ significantly from the geocentric axial dipole field, and these magnetizations are evidently Brunhes-age overprints.

Sites outside the Merritt basin (01, 02, 03, and 25) and in Quilchena Creek valley all have N magnetizations. Individual specimen and site directions are plotted as diamonds in Fig. 9.

Paleomagnetic data are listed by sites in Table 2, with designated magnetization type and appropriate statistics. Average directions and paleopoles are listed in Table 3. In calculating

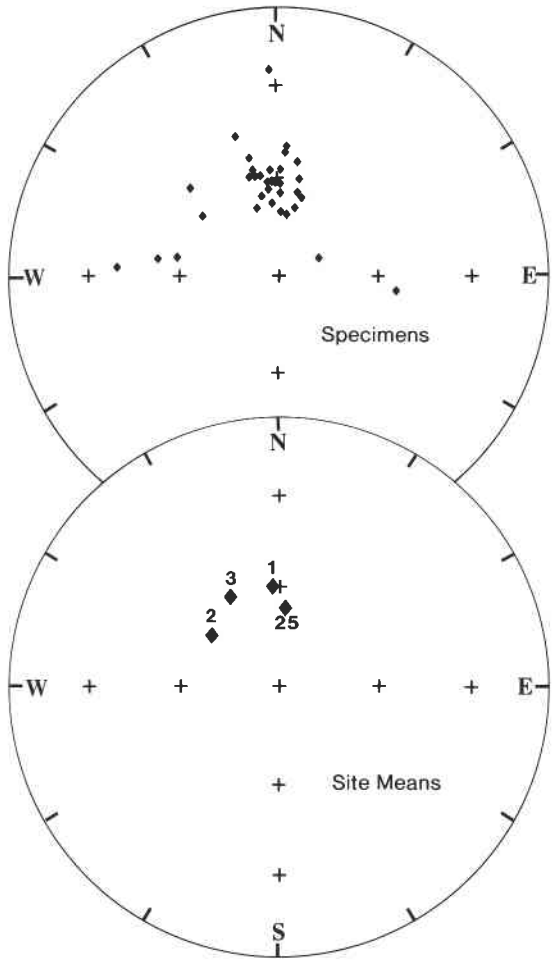


FIG. 9. Directions of magnetization (N) at sites outside the Merritt Basin (01, 02, 03, and 25 of Table 2).

the average of the N magnetizations (entry 2), data from sites 44 and 45, which have anomalously shallow inclinations, were excluded (see discussion near end of paper). The mean of the Brunhes magnetizations (N, entry 2) and of the Brunhes-age overprints (N_0, entry 1) agree with the geocentric axial dipole (GAD) field, the agreement in the former case being especially close (the difference is only 0.4°). Because the sites span time of the order of 10^5 years, this agreement is to be expected. Entry 3 summarizes magnetizations of Matuyama age. There are only four sites, insufficient to average out the secular variation. The mean inclination is about 10° lower than the GAD field. This may have resulted from incomplete removal of Brunhes overprints or may be a consequence of the small number of sites involved.

Stratigraphic interpretation and discussion

By integrating stratigraphic and magnetic polarity data it is possible to outline the events recorded in the succession of Quaternary deposits at Merritt (Table 4). Several alternate interpretations are possible, but the sequence proposed in Table 4 seems to best fit the data.

Reversely magnetized units

The Coldwater silts and sub-Coutlee silts are reversely magnetized and the Coutlee sediments are reversed to within a few metres of their top (see Figs. 3, 5). We interpret these

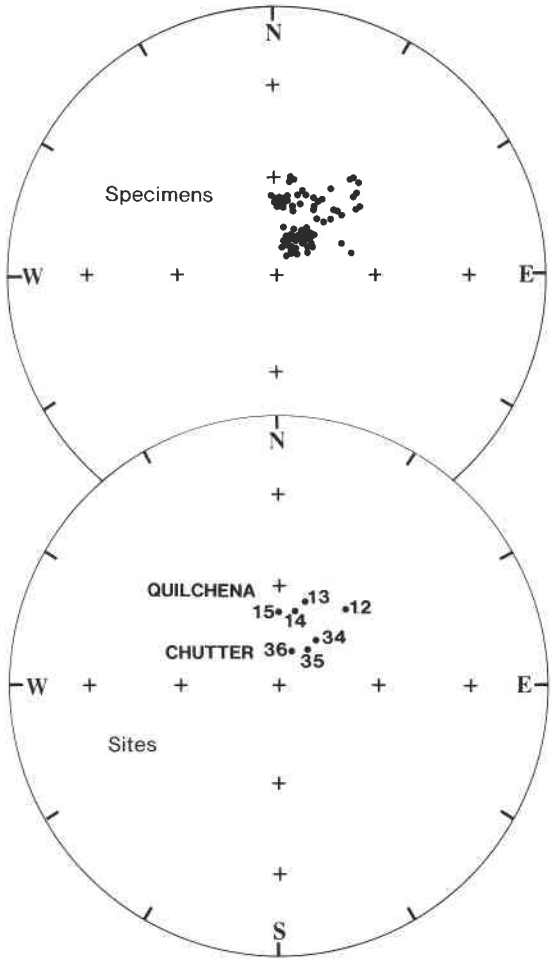


FIG. 10. Specimens (above) and mean lava flow (below) directions from valley basalts at the Quilchena Creek valley and Chutter Ranch localities.

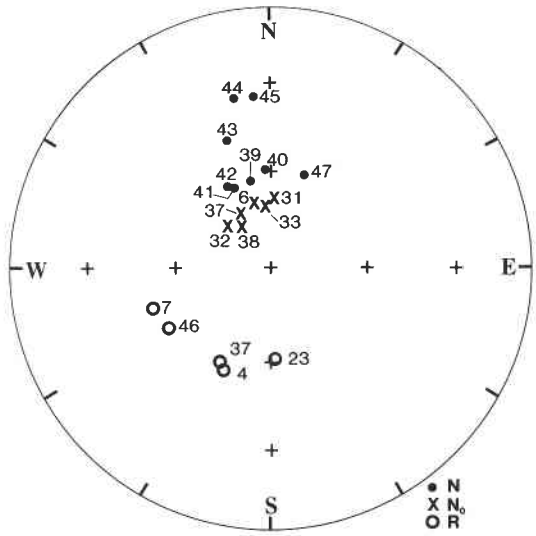


FIG. 11. Site mean directions calculated from sediments of the Merritt basin. N, normal polarity Brunhes magnetizations; N_o, Brunhes overprints; R, reversed magnetizations.

units as belonging to the Matuyama Reversed Polarity Zone (>790 ka) for the following reasons. First, they contain incontrovertible evidence of glaciation, which they would not if they belonged to an earlier reversed polarity zone. (The next oldest reversed zone, the Gilbert, 5−3.4 Ma, is much older than the oldest continental glaciation in North America). Second, these reversely magnetized beds are everywhere overlain by normally magnetized beds for which there are strong grounds for assigning to the Brunhes Normal Polarity Zone (see below). Third, the beds contain the Coutlee tephra, which has yielded a minimum age of >670 ka (Berger 1985), consistent with a Matuyama age.

Coldwater and sub-Coutlee silts

Coldwater and sub-Coutlee silts both have reversed polarity records. Hence both were deposited more than 790 ka ago. Both are glacial lake sediments, and as the two are exposed at the base of exposures only 400 m apart, the simplest explanation is that they were deposited in the same glacial lake. However, the sub-Coutlee silts resemble glacial lake sediments deposited during ice advance, whereas the Coldwater silts are similar to glacial lake deposits laid down during ice retreat. If the two were deposited during the same glaciation with sub-Coutlee silts deposited during advance and Coldwater silts during retreat, Coldwater silts should overlie sub-Coutlee silts at the Lily Lake Road locality. They do not. Another factor suggesting that the two glacial lake units are not closely related in time is that at the Lily Lake Road locality, sub-Coutlee silts are overlain by Coutlee sediments, a unit which is reversely magnetized at the base and normally magnetized at the top; in contrast, at the Coldwater locality, normally magnetized brown drift overlies reversed Coldwater silts. Even if the two silts were deposited in the same glacial lake, the difference in the overlying sequences indicates that subsequent events at the two adjacent localities were different. A further feature that distinguishes the Coldwater and sub-Coutlee silts is their differing declinations (compare sites 07 and 46 with site 04 in Fig. 11). Site 04, along with sites 23 and 37, is directed to the south as expected for an axial field, whereas sites 07 and 46 are rotated clockwise over 40°. This difference is at the limit of what would be expected from secular variation and could be indicative of a time difference between the sites. Alternatively, the data could be taken to imply a substantial clockwise rotation of the Coldwater silts, perhaps as a result of glacial tectonism.

Three possible, relatively simple, interpretations are proposed:

Interpretation 1—Sub-Coutlee silts are older than the Coldwater silts. According to this hypothesis, both silts could have been deposited during the same glaciation, with the sub-Coutlee silts laid down during advance and the Coldwater silts deposited during retreat, or the sub-Coutlee silts could have been deposited during a glaciation earlier than the one responsible for the Coldwater silts. In either case, a period of soil formation followed deposition of the Coldwater silts. Soil formation would have to have been followed by a period of erosion during which the paleosol and the Coldwater silts were stripped only from sub-Coutlee silts at the Lily Lake Road locality prior to the deposition of the Coutlee sediments. Because of the juxtaposition of the two localities, both sites would have to have been covered by Coutlee sediments during the later period of basin filling. Hence, a later period of erosion would be required to remove Coutlee sediments from only the Coldwater locality prior to the deposition of the

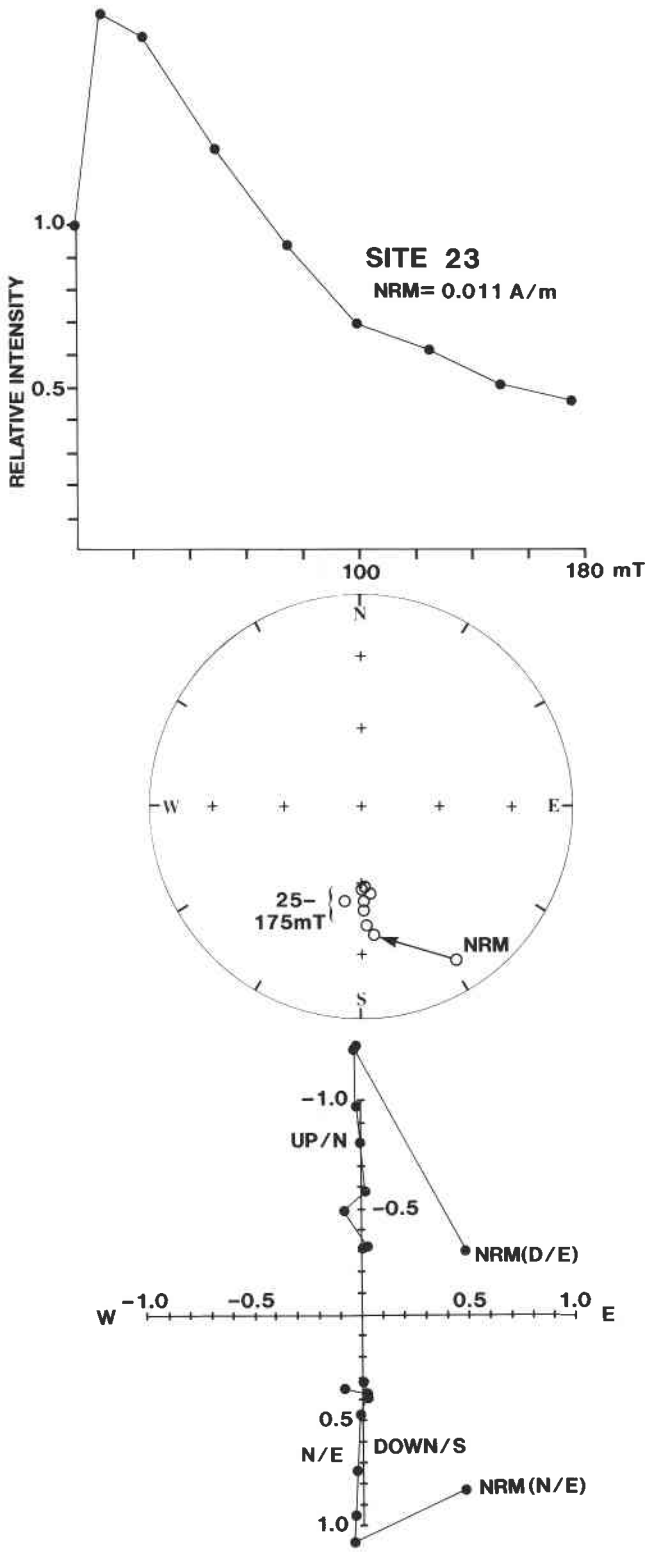


FIG. 12. Reversed (*R*) magnetization in red-brown paleosol at the top of the Coutlee sediments, Lily Lake Road locality, site 23. The beds are horizontal. See Fig. 7 caption.

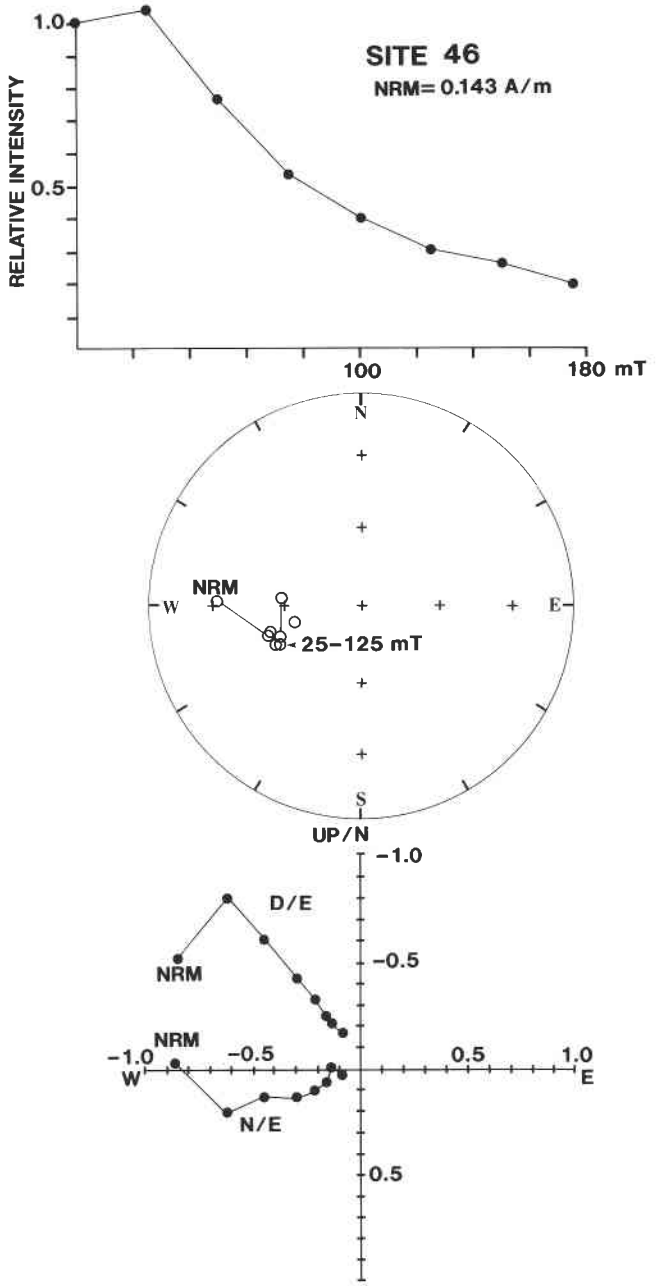


FIG. 13. Reversed (*R*) magnetization in grey glacial lacustrine silt (Coldwater silts) at site 46, Coldwater locality. Directions have been corrected for tilt of bedding (16° at 325° E of north).

normally magnetized brown drift. This interpretation requires one period of erosion that affected the Lily Lake Road locality but not the Coldwater locality and another period of erosion that affected the Coldwater locality but not the adjacent Lily Lake Road one.

Interpretation 2—Sub-Coutlee silts and the Coldwater silts are the same age. If the sub-Coutlee silts and the Coldwater silts are part of the same unit, the only way to explain the presence of a paleosol on top of the Coldwater silts and its absence on top of the sub-Coutlee silts together with the presence of Coutlee sediments on top of sub-Coutlee silts and their absence on top of Coldwater silts is to have two periods of erosion that affected the one locality but not the adjacent one. Under this interpretation, the period of soil formation that followed deposition of these coeval units would have produced soils at the two sites. Erosion limited to the Lily Lake Road locality would then be required to strip this paleosol from the Lily Lake Road locality before the deposition of the Coutlee sediments. Deposition of the Coutlee sediments valley fill then

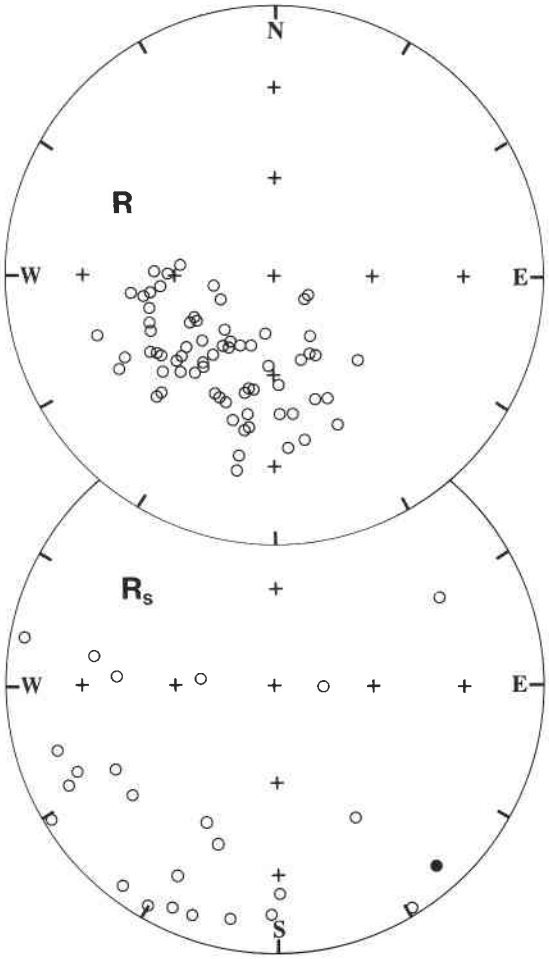


FIG. 14. Reversed magnetizations at the Lily Lake Road and Coldwater localities. Above, reversed (R) magnetizations including the reversed magnetizations at site 37, after removal of small coercive force Brunhes-age over prints. Below, scattered reversed (R_s) magnetizations of small magnitudes that occur in specimens with large, tenacious Brunhes-age overprints. The solid circle denotes negative inclination, the open ones, positive inclination.

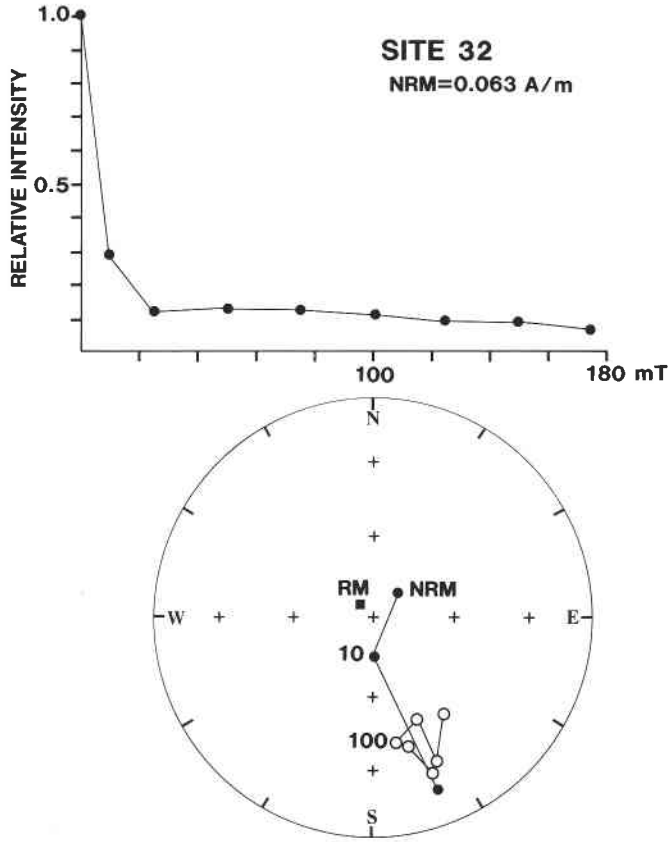


FIG. 15. Reversed (R_s) magnetization in brown silt of the Coutlee sediments, site 32, Lily Lake Road locality. The beds are horizontal. See Fig. 7 caption.

occurred, and was followed by removal of the Coutlee sediments from the Coldwater locality only, prior to deposition of the brown drift.

Interpretation 3—Coldwater silts are older than the sub-Coutlee silts. According to this hypothesis, the paleosol would have been formed during the interglaciation that followed the deposition of the Coldwater silts. The sub-Coutlee silts would have been deposited during a subsequent glaciation, which was followed by the deposition of the interglacial Coutlee sediments. A later glaciation, possibly the one that deposited the brown drift, eroded the valley fill from the Coldwater locality.

All three interpretations require that there were unique periods of erosion affecting one locality but not the other. Interpretation 3, in which the Coldwater silts are older than the sub-Coutlee silts, requires that deposition of the Coldwater and sub-Coutlee silts be separated by an interglaciation (during which a paleosol formed), but it requires only one period of erosion that affected one locality but not the adjacent one. Because it is the least complex, this third interpretation is selected as the preferred one (Table 4).

The casual observer might entertain the possibility that the paleosols at the Coldwater and Lily Lake Road localities are correlative. This would require the unlikely circumstance that either the <40 m thick Coutlee sediments in which the Coutlee paleosol complex developed were never deposited at the Coldwater locality or that they were eroded prior to paleosol formation at that site. Differences in conditions of soil formation and the relative positions of the two sites rule out the possibility that the two paleosols developed at the same time. The paleosol *within* the Coutlee sediments is a complex that developed on several fine-grained colluvial, alluvial, lacustrine, and eolian units. This suggests a flat, poorly drained site where periods of soil formation alternated with periods of sediment deposition of the type one would expect at the margin of a floodplain. The paleosol at the Coldwater locality developed on an erosion surface that truncates the glacial lacustrine Coldwater silts. If the Lily Lake Road locality was lower and lay nearer the centre of the valley than the Coldwater locality, it might be argued that the Coutlee paleosol developed in a floodplain environment at the same time as the Coldwater paleosol was forming on the adjacent valley wall under a stable to slightly erosive environment. This, however, could not have been the case, as the Coldwater site is lower by about 70 m and lies nearer the centre of the valley.

An additional possibility is that the Coldwater silts are an allochthonous glacial tectonic unit. The apparently rotated declination of the remanent magnetization of this unit could be used as evidence for this interpretation (Figs. 11, 13). If the Coldwater silts are allochthonous, they could be no younger than the sub-Coutlee silts because the reversely magnetized

TABLE 3. Mean summary of paleomagnetic directions and paleopoles

Entry no.	Magnetization type	Chronozone	B	D_m (°)	I_m (°)	k	α_{95} (°)	λ (°)	ϕ (°)	K	A_{95}
1	N_o	Brunhes overprint	6	34.1	71.8	158	5.4	76.7	186.9	52	9.4
2	N	Brunhes	7	000.4	66.9	38	5.8	86.2	335.3	14	10.0
3	$R + R_s$	Matuyama	5	217.0	−57.8	24	16.0	62.1	340.0	12	23.3

NOTES: Data obtained by combining entries labelled N_o, N, R, and R_s of Table 2. Entry 1 is the average of sites 06, 31−33, 37, and 38. Entry 2 is the average of sites 01−03, 12−15, 25, 34−36, 39−43, and 47; sites 44 and 45 were excluded because of anomalously low inclinations. Entry 3 is the average of sites 04, 07, 23, 37, and 46. B is the number of collecting sites. D_m and I_m are declination and inclination of the mean; λ and ϕ are the latitude and longitude of the paleopole; α_{95} and A_{95} are the radius of the circles of confidence around the mean direction and the paleopole, respectively; k and K are the precision of site directions and their corresponding virtual geomagnetic poles. The mean location of the sampling sites is 50.1°N, 120.8°W.

TABLE 4. Proposed correlation of Quaternary units in Merritt area

Age (ka)	Event	Unit	Polarity	Occurrence
10 to ~23	Fraser Glaciation	Kamloops Lake Drift		
		Merritt silts	N	Merritt, Lily Lake Road, Coldwater
		Till		Lily Lake Road, Coldwater
		Proglacial sediments	N	Lily Lake Road
>23	Erosion interval	Unconformity		Lily Lake Road
	Glaciation	Boulders on unconformity		Lily Lake Road
		Brown drift(?)	N	Coldwater
	Basalt eruption	Valley basalt	N	Chutter Ranch, Quilchena Creek
~790 to >790	Interglaciation	Uppermost Coutlee sediments	N	Lily Lake Road
	Soil formation	Paleosol top Coutlee sediments	R	Lily Lake Road
	Valley filling	Coutlee sediments	R	Lily Lake Road
	Glaciation	Sub-Coutlee silts	R	Lily Lake Road
	Interglaciation	Paleosol top Coldwater silts	R	Coldwater
	Glaciation	Coldwater silts	R	Coldwater

sub-Coutlee silts underlie a unit that is normally magnetized at the top (Coutlee sediments). This, apparently conformable, succession suggests that reversed silts similar in nature to the Coldwater silts were not deposited in this area after the deposition of the sub-Coutlee silts.

Coutlee sediments

Coutlee sediments record a nonglacial period of basin filling. The lower parts of the sequence contain channel gravels with abundant rounded pebbles from a variety of sources. This suggests that early in the cycle, stream gradients were relatively steep and pebbles from distant sources were carried into the basin. Upward gradation to finer, more poorly sorted sediment, containing clasts of local lithologies as the only coarse material, suggests a decrease in stream gradients later in the cycle. Sediments near the top of the sequence contain paleosols, colluvium, quiet-water sediment, and loess. This suggests periods of subaerial processes during a time of general base-level stability. Iron oxide, which coats many of the clasts in this unit, is thought to have been derived from soils that were developing on uplands during the basin-filling episode. A change from reversed to normal magnetization occurs near the top of the paleosol complex. This part of the section was not sampled in detail, but it is assumed the change in magnetic polarity in the Lily Lake Road section lies at the Brunhes–Matuyama boundary dated at 790 ka. The nature of sediments, combined with the thickness of the valley fill, the large amount of iron oxide (presumed to have come from erosion of a well-developed soil) disseminated uniformly throughout the deposit, and the complex of paleosols near the top of the unit, all suggest that Coutlee sediments were deposited during an interglacial period.

Normally magnetized units

Valley basalts, brown drift, and Kamloops Lake Drift all have normal polarity. They are assumed to belong to the Brunhes Normal Polarity Zone (790 ka to present) because they can all be assigned with reasonable confidence to the later part of the Quaternary and because they overlie reversely magnetized beds.

Valley basalts

The valley basalts in the Merritt basin are older than the Kamloops Lake Drift because pebbles and boulders of the basalts are present in the drift. Also, striations on the surface of the basalt indicate that this unit was overridden by the glaciers that deposited the Kamloops Lake Drift. Like the Kamloops Lake Drift, the basalts are normally magnetized, and, for the above-mentioned reasons, are assigned to the Brunhes Normal Polarity Zone. The occurrence of boulders of valley basalts in the lag on top of the Coutlee sediments suggests that at least one glaciation occurred between extrusion of the lava and the glaciation responsible for the Kamloops Lake

Drift. The basalts are younger than the Coutlee sediments because Coutlee sediments, according to our interpretation, were deposited near the end of the Matuyama Reversed Polarity Chron, whereas the basalts were deposited during the subsequent Brunhes Normal Polarity Chron.

The magnetism of lavas at both localities D and E shows two features of interest. First, at each locality the mean directions of flows differ by small but significant amounts, and trend systematically from east to west. (The flows are numbered in stratigraphic order in Fig. 10). This is clearest at the Quilchena locality (E) where the direction shift is about 20°. By analogy with historical secular variation, this would indicate that the flows were extruded over hundreds not tens of years. Second, the two trends do not overlap, the inclination at Chutter Ranch being significantly steeper. One possibility is that the sequences have been tilted about 10° relative to one another. Although not impossible, this seems unlikely because at present the flows are within a few degrees of horizontal. It is more likely that the flows at the Chutter Ranch and Quilchena localities are not synchronous.

Brown drift

As mentioned above, the brown drift is interpreted as a glacial deposit. Detailed work was not done on this unit and it was inferred to be a drift largely because the included diamicton has the appearance of a till and because the association of deformed gravel, sand, and silt with diamicton is typical of deposits produced by glaciation. Because its brown colour differs from the grey colour commonly seen in Kamloops Lake Drift, and because it is overlain by a grey drift unit presumed to be Kamloops Lake Drift, the brown drift is regarded as the product of a separate glaciation. Alternatively, the brown drift could be a glacially deformed unit at the base of Kamloops Lake Drift, a subunit of Kamloops Lake Drift, or a nonglacial slump deposit. The brown drift is normally magnetized and hence is younger than the reversely magnetized Coutlee sediments. However, instead of overlying the Coutlee sediments, it sits on the Coldwater silts, which are older than the Coutlee sediments. Because of the proximity of the exposures of Coutlee sediments and Coldwater silts, the former at one time must have overlain the Coldwater silts at the Coldwater locality. We suggest the Coutlee sediments were removed by glaciation, the brown drift is the product of that glaciation, and its colour is due to the incorporation of red-brown Coutlee sediments.

Post-Coutlee – Pre-Kamloops Lake Drift interval

The unconformity at the top of the Coutlee sediments spans the period from slightly later than 790 ka until shortly before 20 ka (the time of the advance of the last ice sheet). The gravel that rests on this unconformity includes large boulders of diverse lithologies. The occurrence of these erratic boulders at the base of a generally well sorted gravel sitting on an unconformity suggests that the basal part of the gravel includes a lag that developed during erosion of at least one till. This indicates that at least one glaciation occurred between the end of deposition of the Coutlee sediments and deposition of the Fraser Glaciation Kamloops Lake Drift. If the interpretation of the brown drift at the Coldwater locality as a separate glacial unit is correct, then it is one of the units that has been removed. In addition, the presence of boulders of valley basalt in this lag suggests that at least one of the glacial deposits that was removed contained erratics of, and hence postdated extrusion of, the valley basalts. This leads to the conclusion that the valley basalt predates at least the penultimate glaciation, as its radiometric age (0.6 ± 0.4 Ma) suggests.

The boulder lag on the post-Coutlee unconformity lies at the base of the lower stratified unit of Kamloops Lake Drift (Fig. 3). This indicates that the final stage of fluvial erosion, during which the unconformity was cut to the level at which it is seen today, occurred immediately prior to the last glaciation. Also, the unconformity is exposed at an elevation of about 780 m, almost 165 m above the present floor of the Coldwater River valley. Pebble imbrication in gravels of the lower stratified unit of Kamloops Drift indicates that stream flow was from north towards the south. This leads to the interpretation that at the beginning of the last glaciation, base levels in the Merritt basin were sufficiently high to allow the Merritt basin to drain south via Coldwater River valley instead of westward to the Thompson River as at present.

Kamloops Lake Drift

All three units of the tripartite Kamloops Lake Drift are present in the Merritt basin. The drift is ascribed to the last glaciation, the Fraser Glaciation, which in this area probably occurred ca. 10–20 ka (Fulton and Smith 1978).

The lowest of these three units (unit 6 of Table 1) was deposited during a period of rising base levels, which was caused by increased sediment loads and possibly the blockage of drainage by advancing ice. It is worthwhile noting that the base of this unit is 165 m above present base level and that the lowest sediments are fluvial gravels. If the high base levels had been due to glacial ponding, the first sediments to be deposited on the unconformity would have been lacustrine sediments and not fluvial gravels. Laminated silts do occur higher in the succession and directly underlie till at several localities, indicating that ponding did occur at later times, and immediately prior to ice overriding the area.

During deglaciation several glacial lakes developed in this part of British Columbia (Mathews 1944; Fulton 1969). Because of the shape of the ice front and the timing of ice retreat, glacial Lake Hamilton and glacial Lake Merritt probably were the only glacial lakes to occupy the Merritt basin. The elevation of the Merritt silts at the top of the Lily Lake Road locality indicates that these sediments were deposited in glacial Lake Hamilton. Silts at the Merritt locality are in a position such that they could have been deposited in both glacial lakes Hamilton and Merritt. A varve count at the Merritt locality indicates that glacial lake sedimentation occurred over at least 150 years.

Sites outside Merritt basin

The sediments sampled at localities outside the Merritt basin are all of different ages. South Thompson silt at Barnhart Vale is related to the end of Fraser Glaciation; the clay at Mission Flats is part of a succession containing Middle Wisconsinan tephras; the varved silt and clay at Okanagan Centre underlies Middle Wisconsinan deposits and is related to the end of the penultimate glaciation; and the marl at Westwold underlies glacial deposits that are at least as old as the penultimate glaciation, hence are at least as old as the Sangamonian. Stratigraphic considerations indicate that deposits at the first three sites are younger than 790 ka, and they are normally magnetized as expected. However, the age of the marl at Westwold was less certain, but its normal polarity indicates that it, too, is younger than the Brunhes–Matuyama reversal.

A point of note is that the Merritt silts at Merritt and the South Thompson silt at the Barnhart Vale were both deposited

during retreat of Fraser Glaciation ice ca. 10–11 ka. The paleodeclinations at the two localities are similar but the Merritt silts have a lower inclination than the South Thompson silt and indeed a shallower inclination than most Brunhes National Polarity Chron deposits (Table 2, sites 44, 45, and 01). The low inclination of the Merritt silts might be attributed to rotation of grains during compaction of the lacustrine sediment, but the South Thompson silt samples were buried by a comparable thickness of sediment and show no evidence of similar rotation of the paleomagnetic orientation. It might be speculated that the difference was caused by different drainage histories of the two lakes. Possibly, the South Thompson silt was deposited in a glacial lake that drained quickly, soon after deposition of the sediment and with little compaction (Evans and Buchanan 1976, remark on anomalously high void ratios in the South Thompson silt), whereas the Merritt silts were deposited in a lake where delayed drainage permitted some compaction and dewatering.

Conclusions

1. Quarternary deposits in the Merritt area provide evidence for at least three glaciations, and if the interpretation of Coldwater and sub-Coutlee silts as resulting from different glaciations is correct, for four (Table 4).

2. Three distinct paleomagnetic signatures are present: normal polarity magnetization acquired during the Brunhes Normal Polarity Chron, reversed polarity acquired during the Matuyama Reversed Polarity Chron, and reversed polarity (Matuyama) magnetization all but obscured by a normal polarity overprint (Brunhes).

3. A succession of reversely magnetized deposits, which includes nonglacial sediments, paleosols, and glacial lacustrine sediments, is interpreted as including deposits and evidence of at least two glaciations and two interglaciations that occurred prior to the Brunhes–Matuyama magnetic reversal (prior to 790 ka).

4. The normally magnetized sediments at Merritt contain evidence for the end of an interglacial, an extrusion of basalt, a glaciation, and a period of erosion prior to the last glaciation.

5. Valley basalt was extruded over a period of hundreds of years in the Merritt basin after 790 ka but prior to the penultimate glaciation. Similarly appearing basalts were extruded in Quilchena Creek Valley after 790 ka and the position of the paleopoles for the two volcanic units suggests they probably were extruded at different times.

6. Base levels in the Merritt basin were high immediately prior to the last glaciation, and the drainage was to the south rather than to the west to the Thompson River.

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Appendix

When magnetizations were simple single components, as for instance in the valley basalts, the direction of the principal magnetization was obtained by choosing a single AF demagnetization level based on the behaviour of pilot specimens. For most samples of sediments, especially those containing reversed magnetizations, demagnetizations were carried out in 5–10 steps and magnetizations were determined by line fitting. Except at site 37, R_s magnetizations were small but very resistant to demagnetization, and line fitting was not possible; they were determined from crude end points observed after demagnetization in very high alternating fields. Thermal demagnetization of valley basalt sites 12–15 revealed unblocking temperatures mainly in the range 500–600°C. Data from four specimens treated at 500°C were included, bringing the total to 37. The details for each site are as follows:

Site	AF (mT)	s	Site	AF (mT)	s
01	80	8	25	20	15
02	10	3	31 N_o	10–50	6
03	30	8	31 R_s	150	3
04	100	9	32 N_o	10–175	4
	175	10	32 R_s	175	4
	30–100	5	33 N_o	10–100	7
	10–175	3	33 R_s	150	2
06 N_o	10–80	4	34–36	50	41
06 R_s	150	10	37 N_o	10–50	4
07	175	12	37 R_s	125	4
	50–175	4	38 N_o	10–100	5
12–15	20	33	38 R_s	175	6
23	100	6	39–41	25–175	19
	150	5	42–45	50–175	26
	50–175	4	46, 47	50–75	13